



Hybrid AI

Neural and Symbolic Reasoning

VL01 · What can neural and symbolic methods achieve together?

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Your first logic program

```
world open.  
socrates:Human.  
socrates[name -> "Socrates"].  
  
?X:Mortal <- ?X:Human.  
  
?- socrates:Mortal.
```

LINE BY LINE

Socrates is a human.

His name is «Socrates».

Every human is mortal.

Is Socrates mortal?

The answer is **true**.

Most of you have never seen a logic syntax before. You could still read it.

Where we are right now

Sessions	What you learn to do	
01	What is Hybrid AI?	 YOU ARE HERE
02 · 03	Represent knowledge, derive consequences, build H1	29 September
04 · 05 · 06 · 07	Graphs, LLM reasoning, grounding, build H2	10 November
08 · 09 · 10	Bounded search, feedback, build H3	1 December
11 · 12	Learning with structure, choosing an architecture	Transfer

Three things you should be able to do

01

SEPARATE

Say what the neural side and the symbolic side each give you, and what each cannot.

02

ARRANGE

Place a hybrid design: before, during or after, and lowering or lifting.

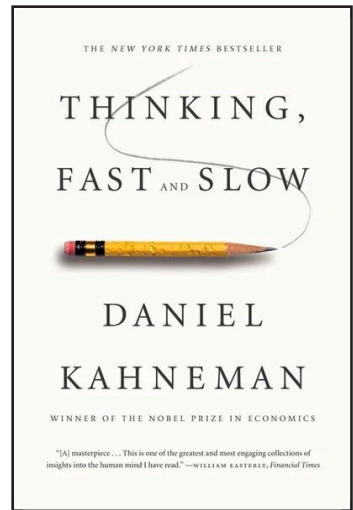
03

DECIDE

For one step of an agent, say who decides, what can refuse, and what counts as done.

Two systems

What each side gives you, and what it cannot



Hybrid AI [1]

Logic

Facts

Reasoning

Rationalisation

Creativity by Deduction

Plan & Adapt
Enact

Analogy

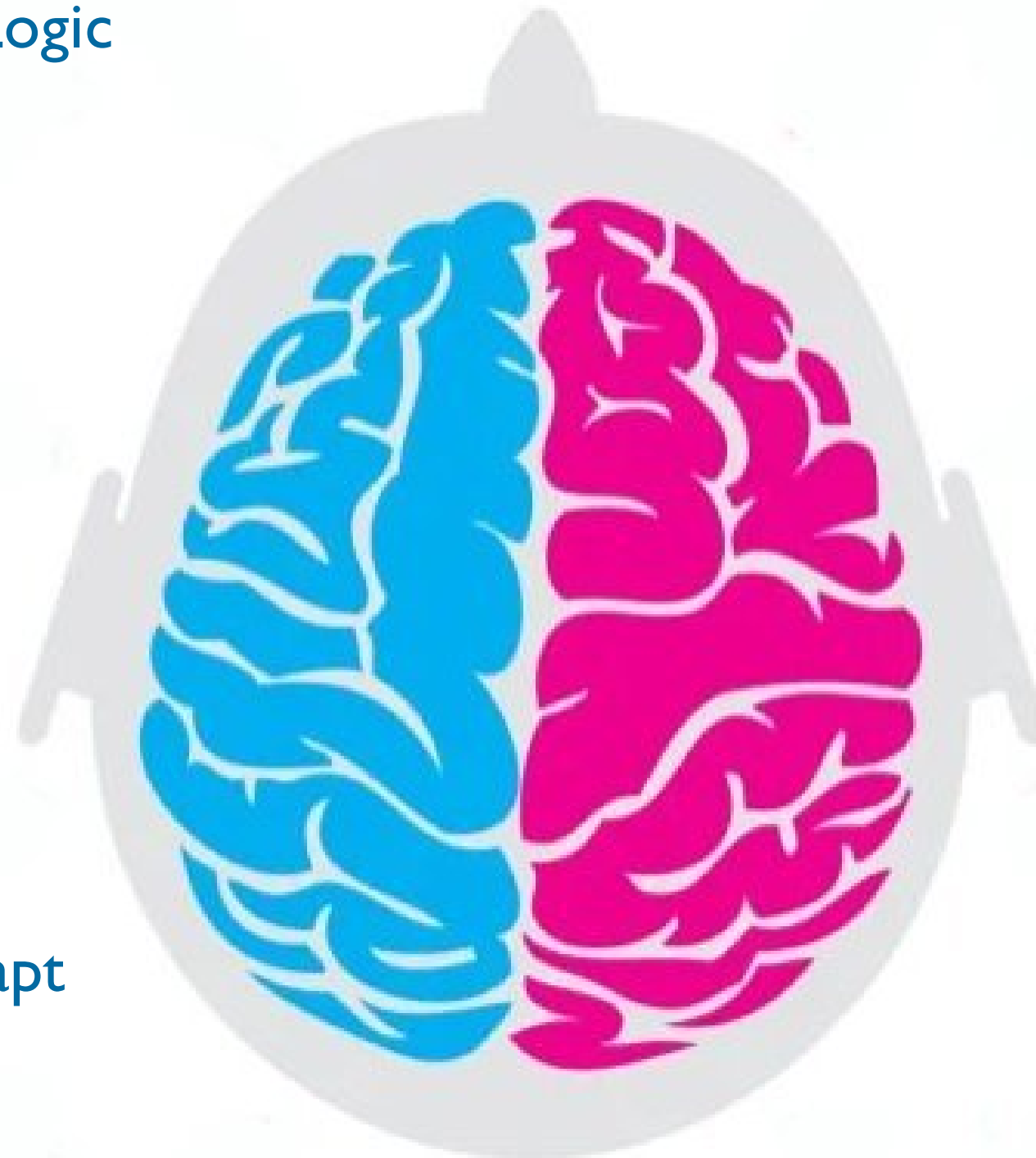
Intuition

Analogical Reasoning

Imagination (Hallucination)

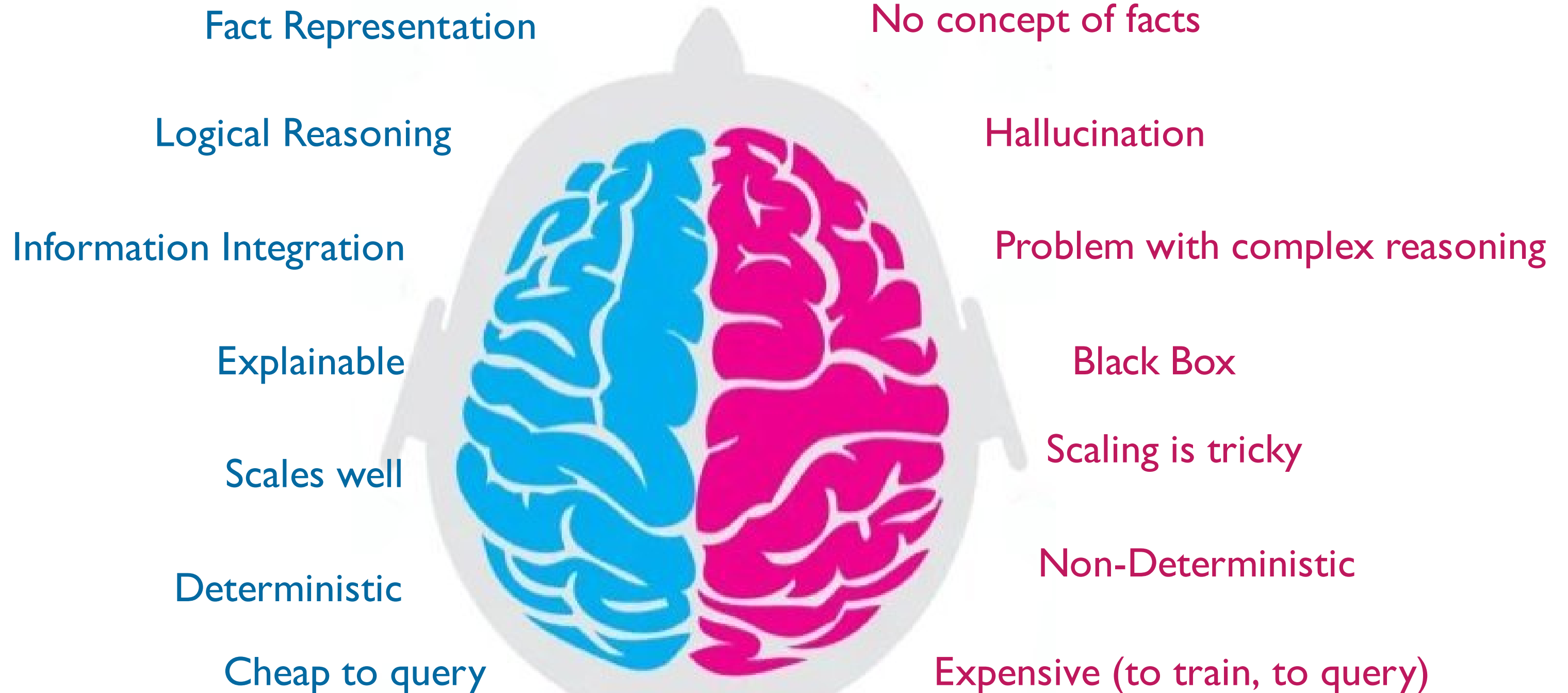
Combinatorial Creativity

Art



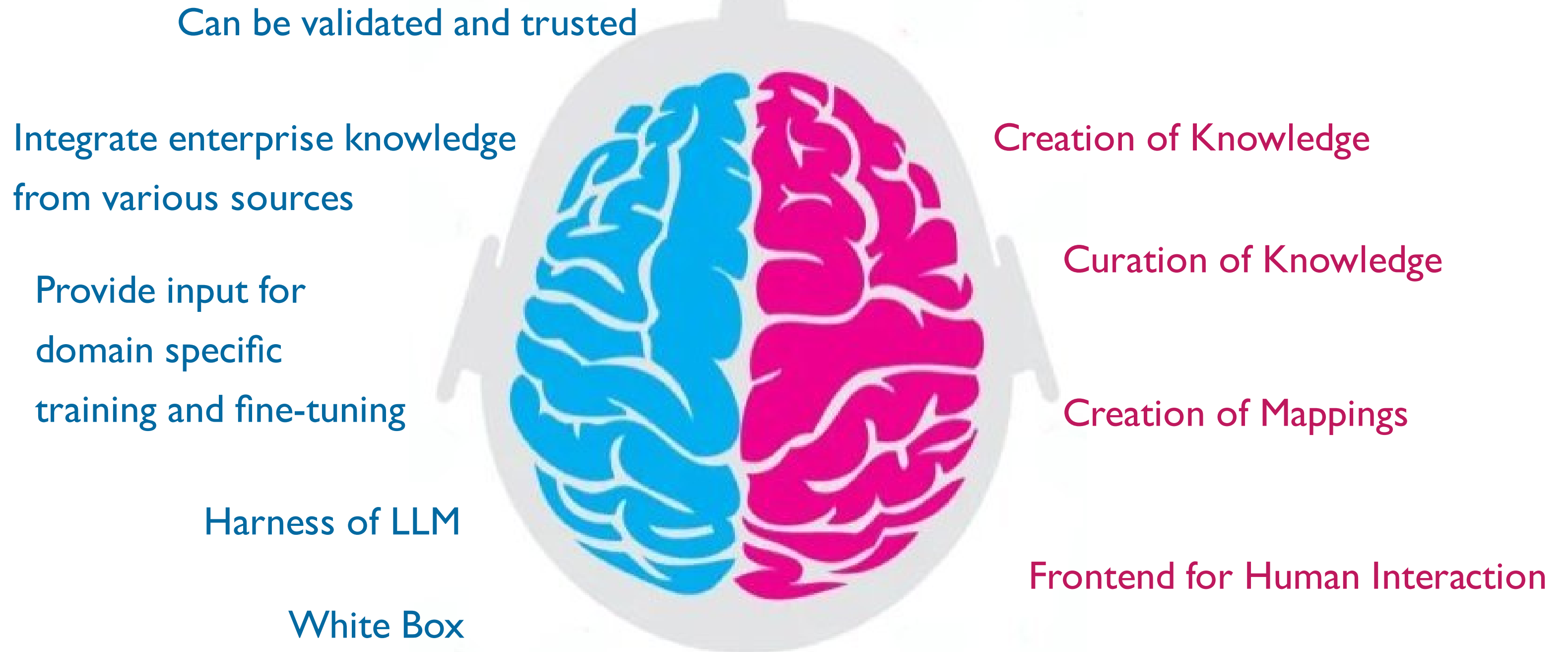
Symbolic System

Large Language Models



Symbolic System

Large Language Models





Take-Home: Two systems

01

TWO KINDS

Explicit knowledge and learned patterns are complementary capabilities, and neither is the whole of intelligence.

02

FACTS

A stored assertion can be inspected and withdrawn. It can still be wrong.

03

FLUENCY

Fluent intermediate steps are not a proof. Plausible is not verified.

04

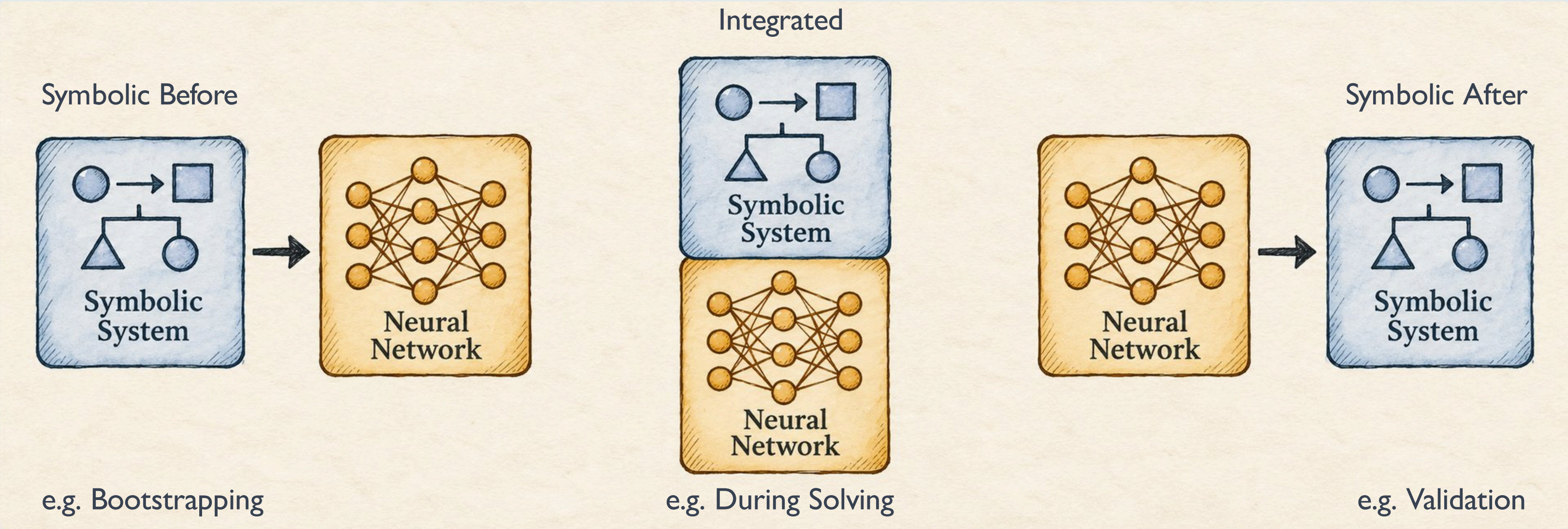
BOTH WAYS

Symbols can guide a model, and a model can help build and curate the symbols.

How they combine

Three places, two directions, four families

Combinations of Neuro and Symbolic [2]



Two Main Categories of Neurosymbolic AI ^[3]

Category 01

LOWERING

Compress symbolic knowledge into neural patterns

The goal is to embed explicit rules and knowledge so that the neural system can reason efficiently using learned representations.

Category 02

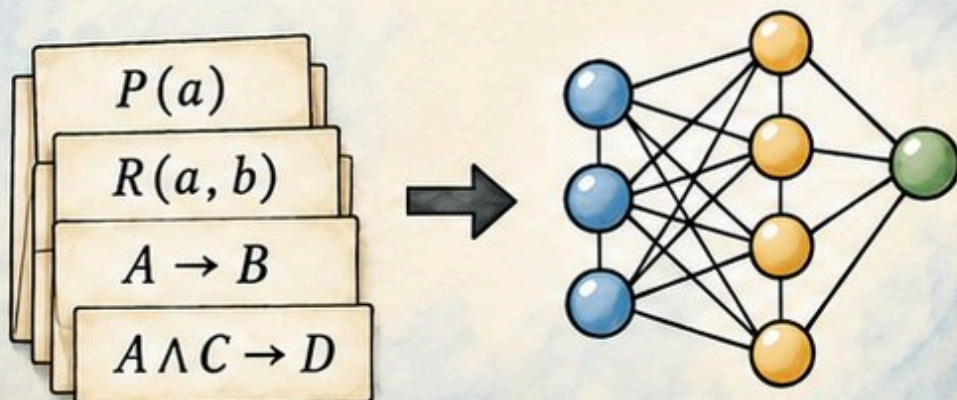
LIFTING

Extract structure from neural patterns into symbolic forms

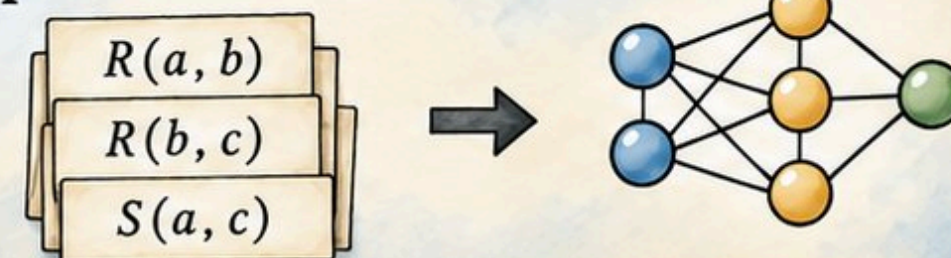
The goal is to map continuous neural outputs into structured knowledge representations to perform rigorous symbolic reasoning.

Neurosymbolic AI

Knowledge into neural models



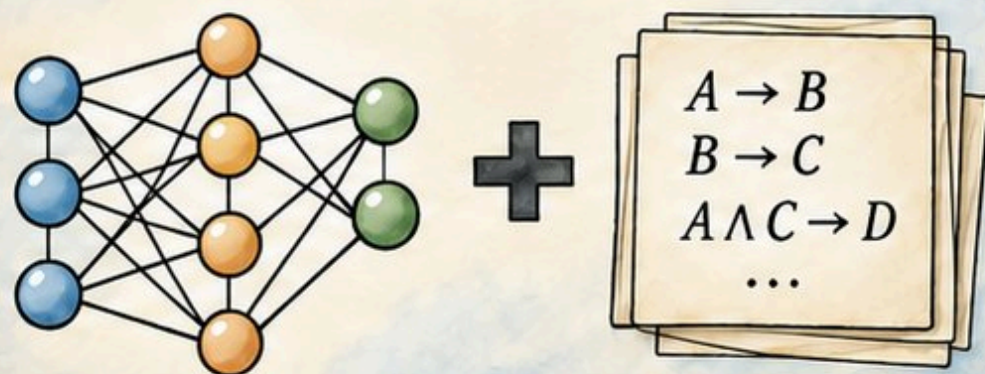
Knowledge graph compression



Logic compression



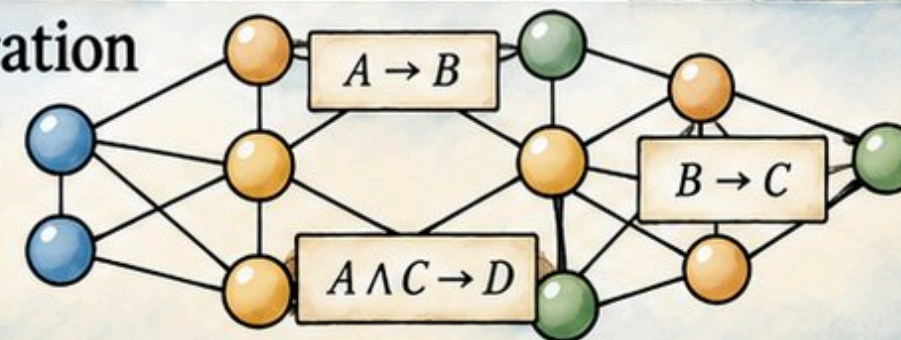
Neural + symbolic reasoning

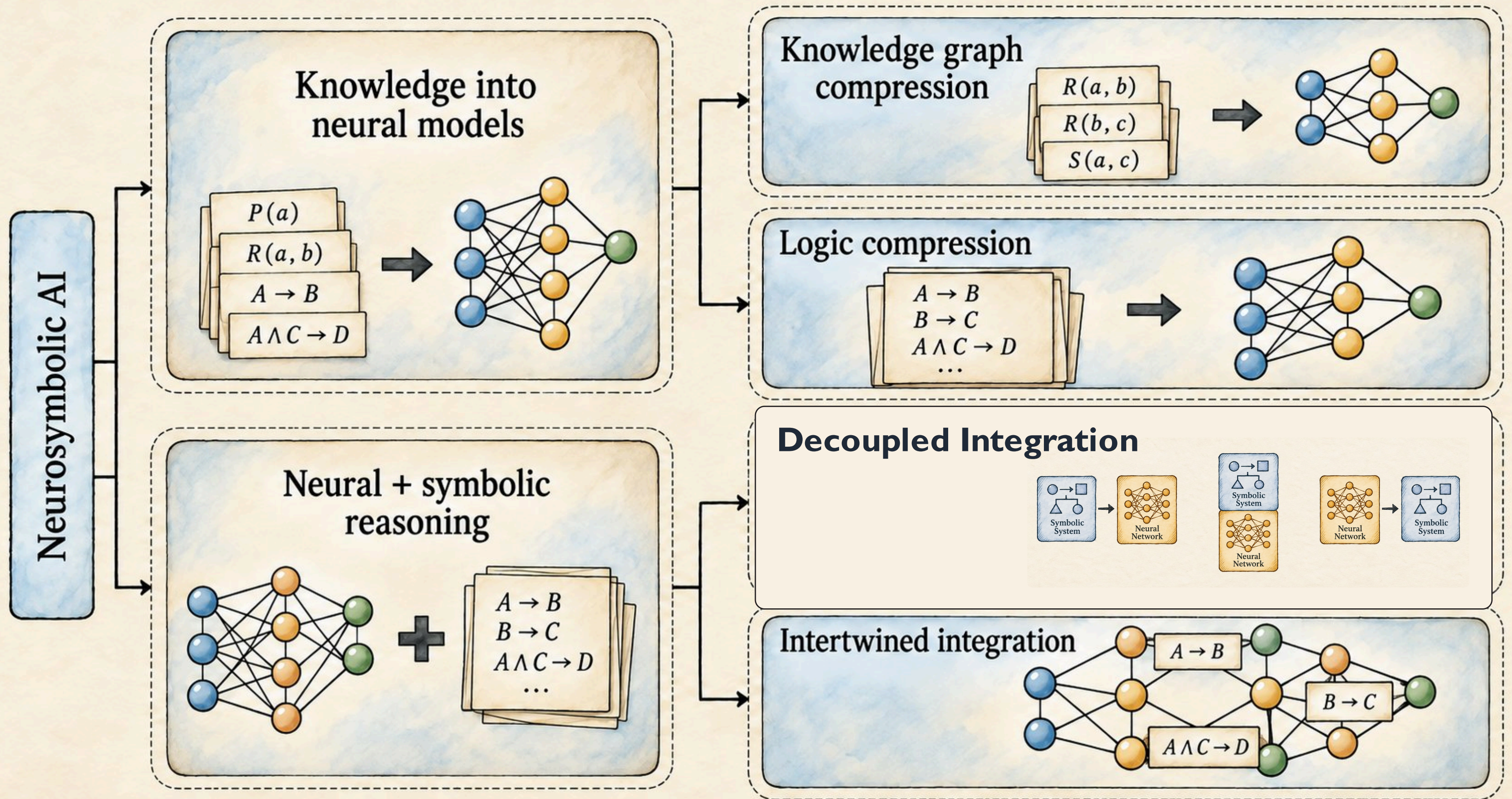


Decoupled integration

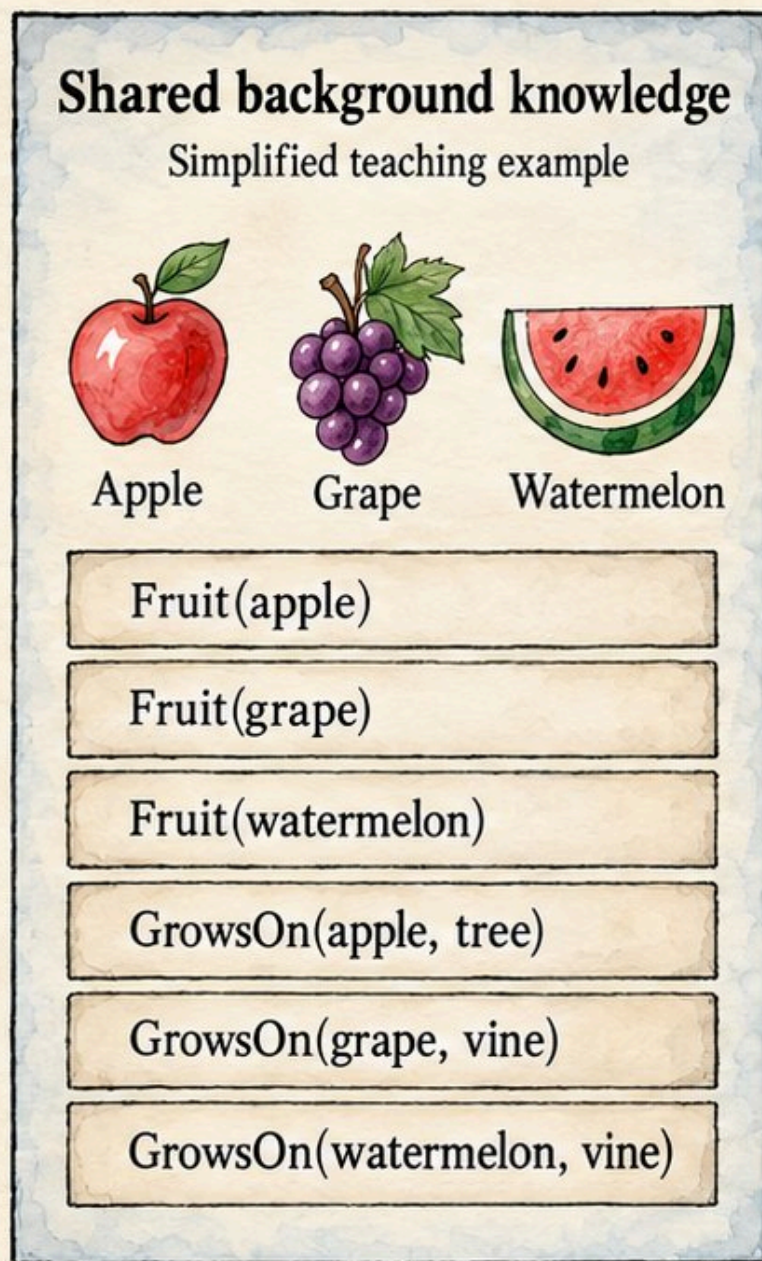


Intertwined integration





Two ways to integrate structured knowledge



Construct
mask

Guide
learning

Knowledge shapes processing

Attention mask

Inductive bias

Masked attention

	Apple	Grape	Watermelon
Apple	1	0	0
Grape	0	1	1
Watermelon	0	1	1

Apply to
attention

	Apple	Grape	Watermelon
Apple		X	X
Grape	X		
Watermelon	X		

Same growth habit or self
1: allowed · 0: blocked

This mask is designed using growth habit knowledge.
Not a universal biological or attention rule.

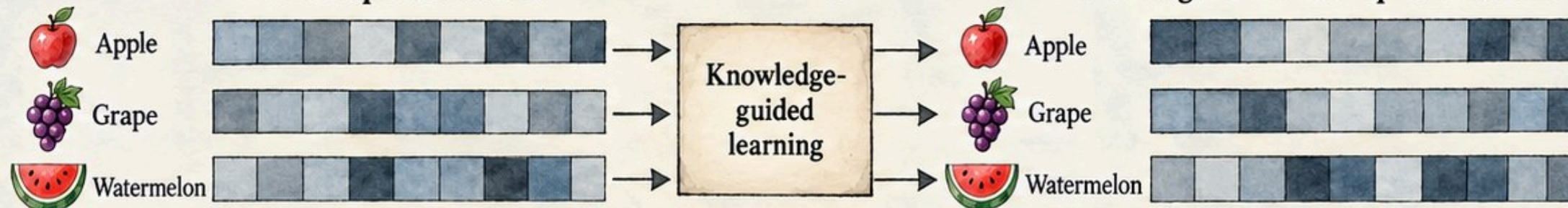
Knowledge constrains which interactions are allowed.

Knowledge shapes representations

Initial representations

Embeddings

Knowledge-enriched representations



Knowledge influences what the vectors encode.

Schematic representations and attention weights; not measured values.

Decoupled Integration

Everything running in production today remains decoupled, even with frequent integration.

Everything else are research prototypes.



Take-Home: How they combine

01

TOPOLOGY

Symbolic logic can precede generation, interleave during inference, or validate final output.

02

ABSTRACTION

Lowering compiles rules into weights, lifting extracts structured assertions back out: scale against trustworthiness, an inherent tension.

03

COUPLING

Architectures range from knowledge-infused weights to decoupled message-passing pipelines.

04

CONSTRAINTS

Hard attention masks enforce their restriction only where they apply.

Four worked examples
Tools, proofs, retrieval, stories

Neural models use external tools

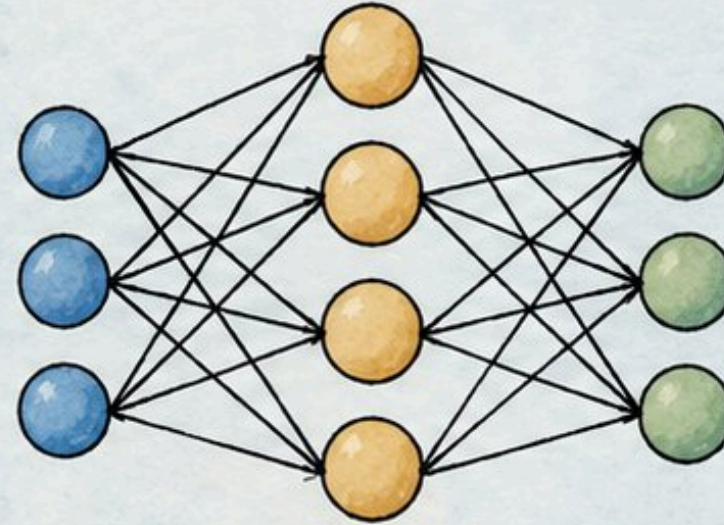
A simple travel-planning example

Input question

How much time should I allow for driving from New York to Los Angeles, including one hour of preparation?



Language model



Selects tools and uses their results

Driving-time query

41 h (illustrative estimate)

1. Retrieve route information



Route service

2. Calculate the total



Calculator

$41 + 1$

42 h

Compose answer

Answer

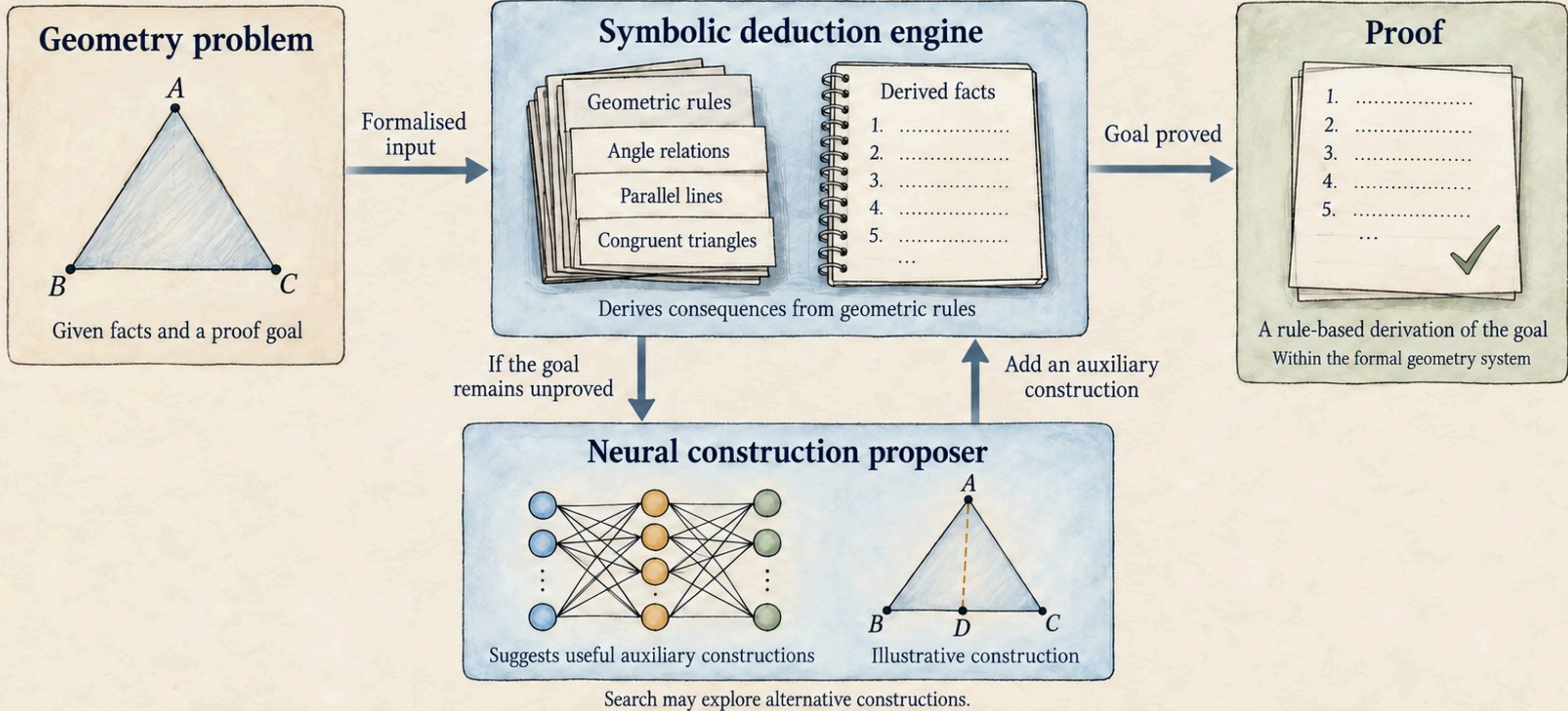
About 42 hours for driving and preparation.

Stops, sleep and traffic delays require additional time.



Neural proposals meet symbolic deduction

AlphaGeometry2 · 2025



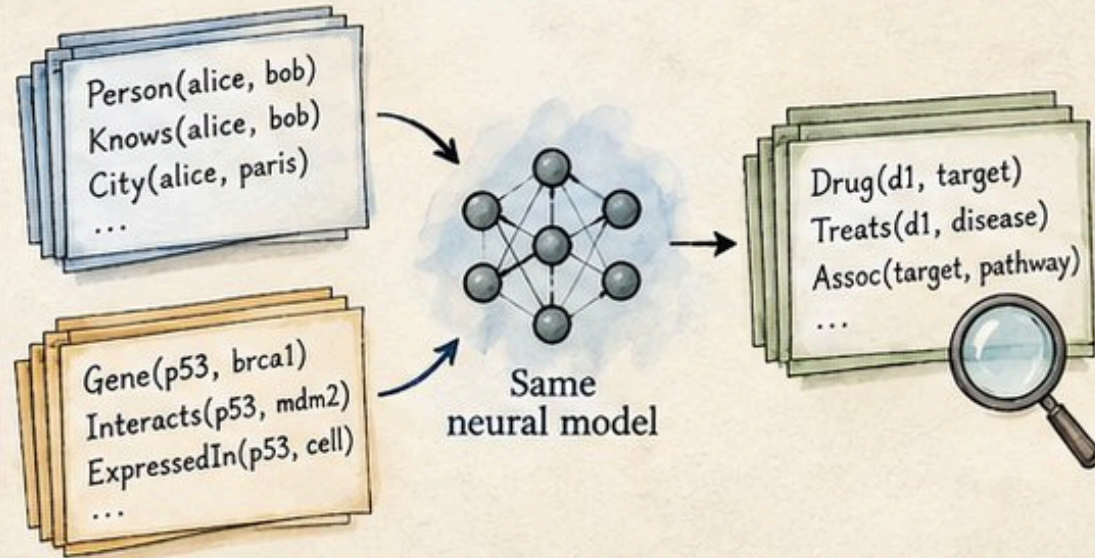
The neural model proposes a way forward; the symbolic engine derives the proof.

Neurosymbolic AI: New Developments

2024–2026 · Selected approaches

Transferable graph representations

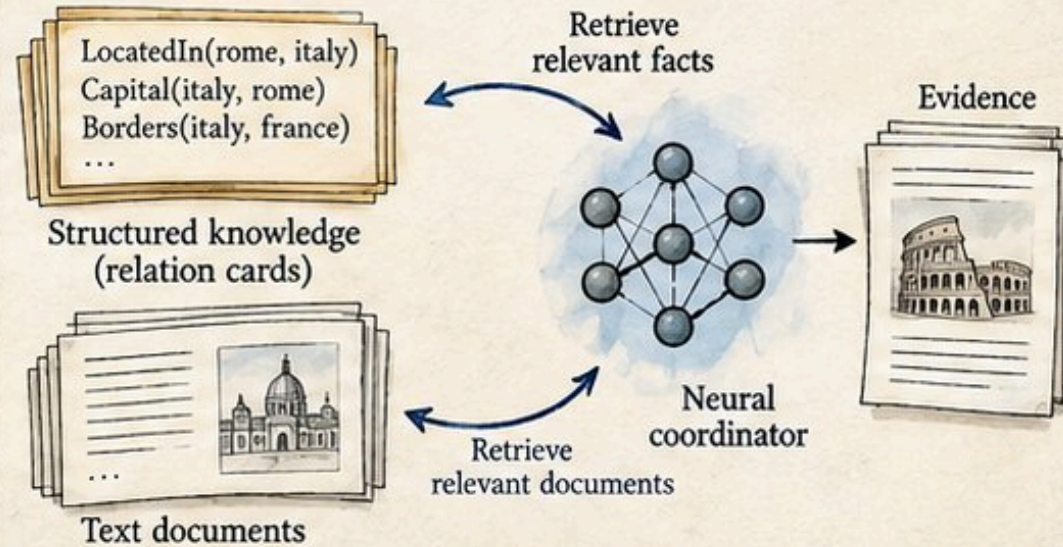
ULTRA · 2024



Transfer learned patterns to unseen graphs

Knowledge-guided retrieval

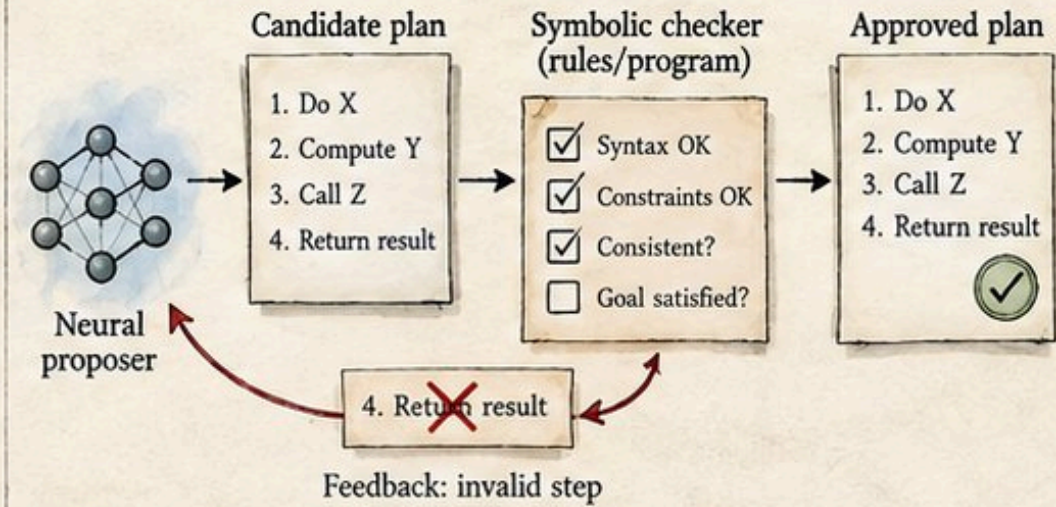
Think-on-Graph 2.0 · 2024



Navigate structured knowledge and text

Verification-guided refinement

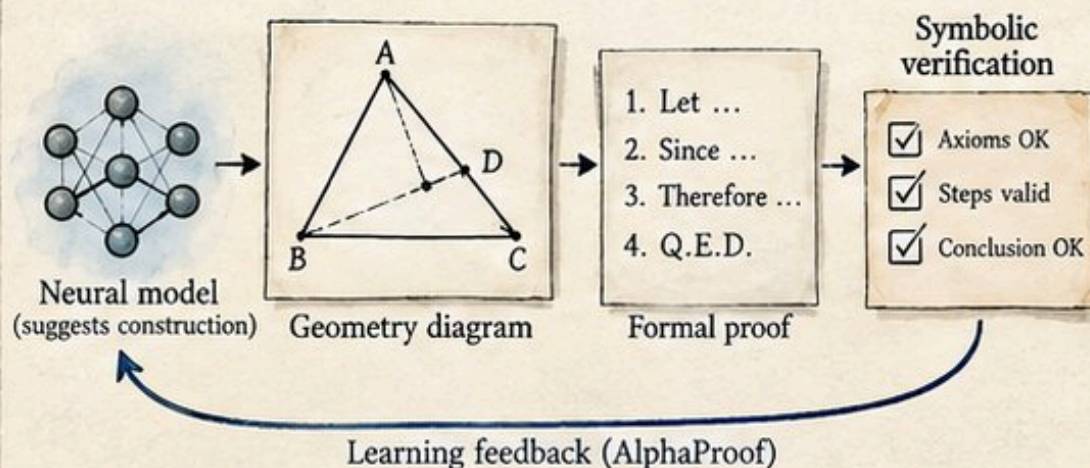
LLM-Modulo · 2024



Propose, check, revise

Neural-guided proof search

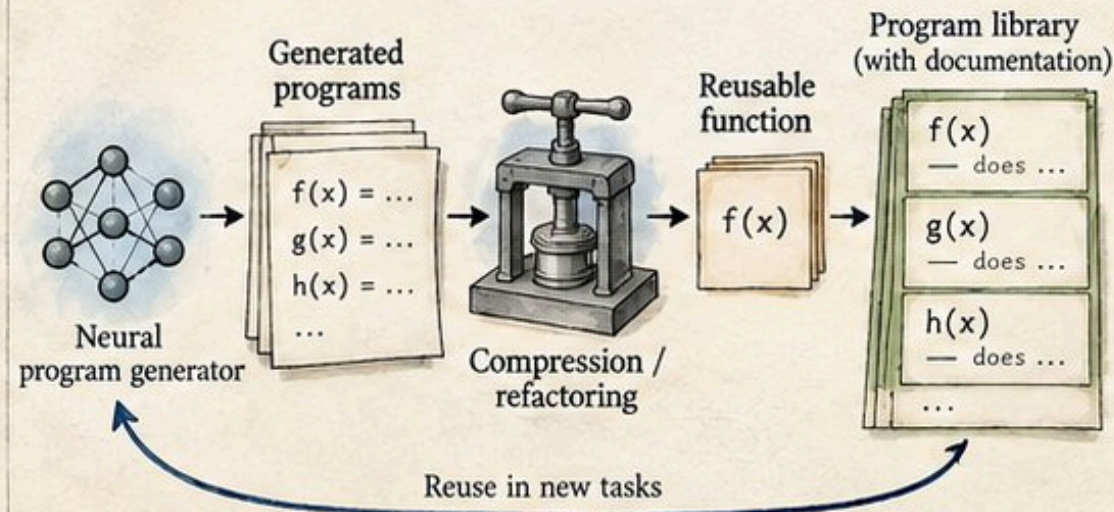
AlphaGeometry2 · 2025 | AlphaProof · 2024/25



Guide symbolic search; verify formal results

Program abstraction learning

LILO · 2024

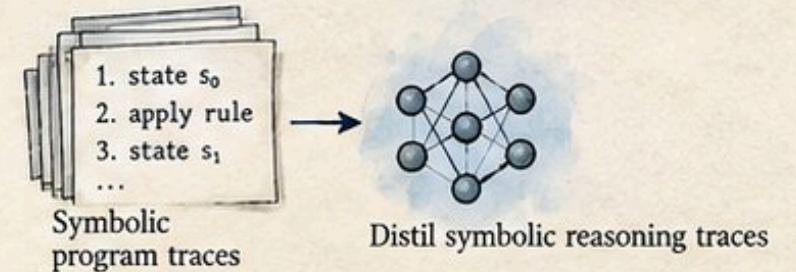


Build and reuse symbolic program libraries

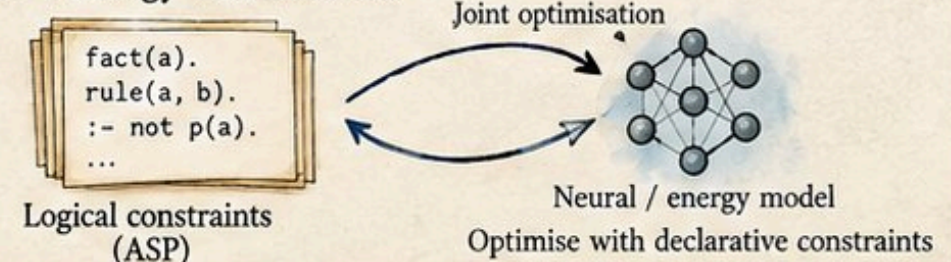
Emerging directions · 2026

Preprints

APEIRIA



ASP + energy-based models



Shape the plot. Check the logic. Tell the story.

A proposed FrameX workflow for controlled story generation

1. Propose an event



LLM

Lea identifies Paul by his handwriting.

?

ARCHIVE

Revise the event plan

2. Check prerequisites

FrameX



Lea has found the note.



Does Lea know Paul's handwriting?

Unknown



Identification requires prior familiarity.



Missing prerequisite: establish familiarity first.

3. Repair the plan

Earlier: Lea studies Paul's signed letter.

Proposed story revision

Later: Lea finds the anonymous note.

Then: Lea recognises the handwriting.

4. Write the scene



LLM

Lea unfolded the note. She recognised the handwriting from Paul's signed letter.

FrameX checks modelled prerequisites. The LLM proposes and narrates.

Logical consistency supports the plot; it does not guarantee a good story.



Take-Home: Worked examples

01

DELEGATION

The model orchestrates and formats; external APIs and calculators evaluate exact values.

02

VERIFICATION

Neural generation suggests candidate paths; formal solvers determine validity.

03

FEEDBACK LOOPS

Iterative refinement pairs generative proposals with rule-based critiques.

04

OPEN WORLD

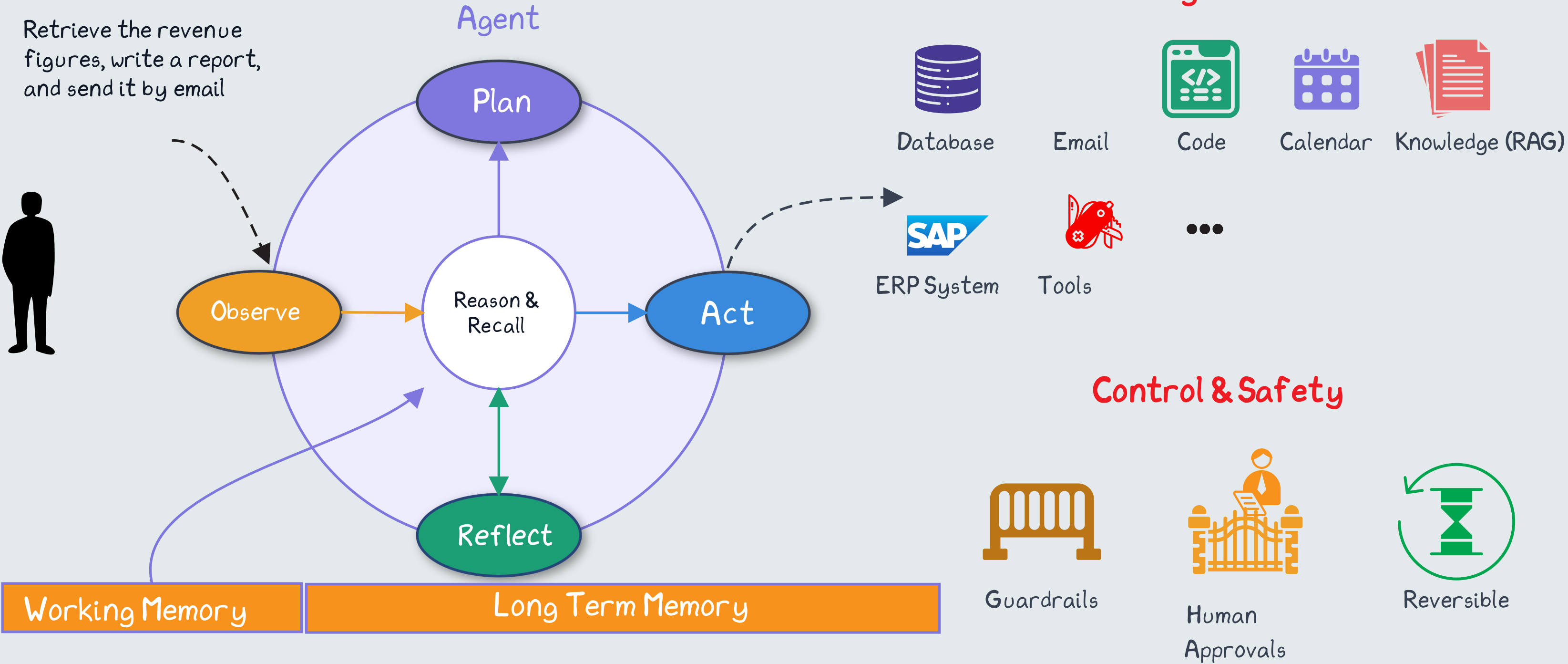
Unproven prerequisites trigger plan repairs, while formal contradictions reject actions.

Agents

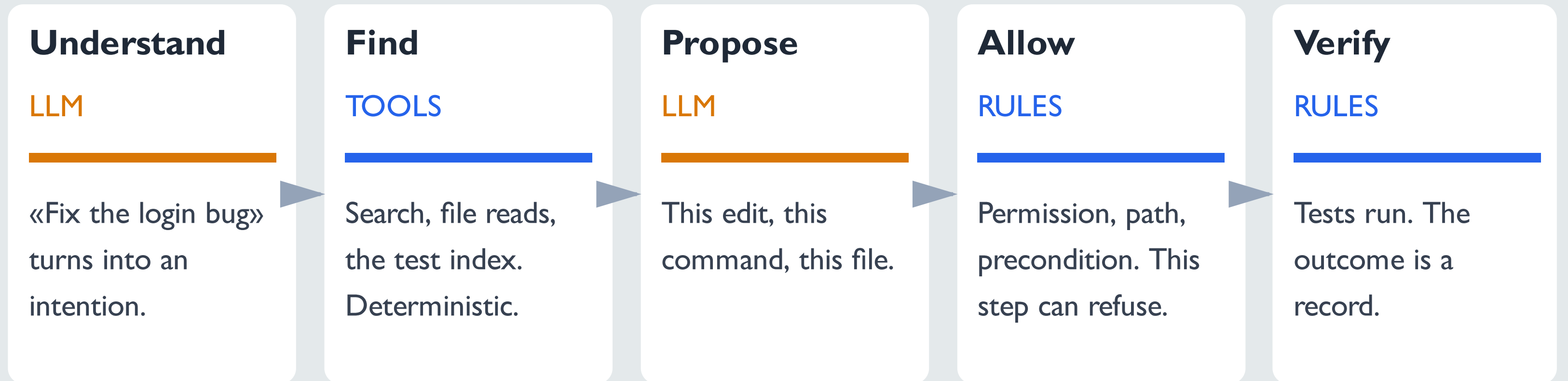
Who decides at each step

Agentic AI

AI that does not just respond, but takes action ^[14]



A coding agent, taken apart



Orange is the model. Blue is everything that runs the same way every time. The loop goes back from **Verify** to **Propose** until the tests are green.

Today most of the blue boxes are permission prompts and hard-coded checks. What if they were rules you could state and reason about?

What about reasoning inside an LLM?

Generate

INTERMEDIATE STEPS

Writing out steps puts computation between the question and the answer.

Train

LEARNED STRATEGY

Training can change which strategies are learned.

Use

SAMPLE AND SELECT

Sampling, selection and feedback change how a model is used.

VL05 takes these apart and asks what the evidence really says.



Take-Home: Agents

01

THE LOOP

Observe, plan, act, reflect. State persists across steps, not inside the prompt.

02

AUTHORITY

The model suggests steps; your host code runs the tools and validates output.

03

GATING

System prompts advise. Hardcoded gates block execution.

04

REASONING

CoT tokens, fine-tuning, and test-time search are distinct engineering knobs.

Summary

Three things you can now do

01

SEPARATE

Distinguish explicit, inspectable facts and rules from probabilistic pattern completion and candidate proposals.

02

ARRANGE

Map any hybrid architecture by pipeline stage (before, during, after) and representation flow (lowering vs. lifting).

03

DECIDE

Deconstruct an agent step into proposal, deterministic gating, external tool execution, and verification.

You started with five lines of unfamiliar logic syntax. Now you know where declarative knowledge fits into neural workflows and how explicit gates enforce boundaries.

Next: VL02 puts FrameX on screen: asserting facts, deriving rules, and handling the open-world assumption.

Kahoot Quiz:Active Recall

[Kahoot Quiz VL01](#)

play.kahoot.it

Working vocabulary

Term	Meaning here
Fact	Something asserted, written <code>object[property → value]</code>
Variable	A name with <code>?</code> , filled in by the engine
Gate	A check that can refuse an action, as opposed to a prompt that asks
Rule	A condition under which a consequence follows, written with <code>←</code>
Inference	Computing consequences from facts and rules
Grounding	Linking a request to the entities and operations you meant
World assumption	What a missing fact lets you conclude
Baseline	A comparison system with a stated information budget

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